Exercise 7.2 SOLUTION

Q1. Evaluate the following:

Formula for permutations ${}^{n}p_{r} = \frac{n!}{(n-r)!}$

i. ${}^{10}P_5$

Sol: ${}^{10}P_5 = \frac{10!}{(10-5)!} = \frac{10.9.8.7.6.5!}{5!} = 30,240$

ii. ${}^{5}P_2$

Sol: ${}^{5}P_2 = \frac{5!}{(5-2)!} = \frac{5.4.3!}{3!} = 20$

iii. ${}^{7}P_7$

Sol: ${}^{7}P_7 = \frac{7!}{(7-7)!} = \frac{7.6.5.4.3.2.1}{1} = 5040$

iv. ${}^{10}P_3$

Sol: ${}^{10}P_3 = \frac{10!}{(10-3)!} = \frac{10.9.8.7!}{7!} = 720$

Q.2. Find the value of n when:

i. ${}^{n}P_3 = 504$

Sol: $\frac{n!}{(n-3)!} = 504$

$\frac{n(n-1)(n-2)(n-3)!}{(n-3)!} = 504$

$n(n-1)(n-2) = 9.8.7 \Rightarrow n = 9$

ii. ${}^{15}P_n = 15.14.13.12.11$

Sol: ${}^{15}P_n = 15.14.13.12.11 \quad (\text{'}\times\text{'} and \div by 10!)$

$\frac{15!}{(15-n)!} = \frac{15.14.13.12.11.10!}{10!}$

$\frac{15!}{(15-n)!} = \frac{15!}{10!} \Rightarrow (15-n)! = 10!$

$\Rightarrow 15-n = 10 \Rightarrow 15-10 = n$

$\Rightarrow n = 5$

iii. ${}^{n}P_5 : {}^{n-2}P_2 = 540 : 1$

Sol: ${}^{n}P_5 : {}^{n-2}P_2 = 540:1$

$\frac{{}^{n}P_5}{{}^{n-2}P_2} = \frac{540}{1}$

$\frac{\frac{n!}{(n-5)!}}{\frac{(n-2)!}{(n-2-2)!}} = \frac{540}{1}$

$\frac{n!}{(n-5)!} \times \frac{(n-4)!}{(n-2)!} = \frac{540}{1}$

$\frac{n(n-1)(n-2)!}{(n-5)!} \times \frac{(n-4)(n-5)!}{(n-2)!} = 540$

$n(n-1)(n-4) = 540$

$n(n-1)(n-4) = 10.9.6 \Rightarrow n = 10$

Q.3. Prove from the first principle that:

i. ${}^{n}P_r = n \cdot {}^{n-1}P_{r-1}$

Sol: R.H.S $= n \cdot {}^{n-1}P_{r-1}$

$= n.\frac{(n-1)!}{(n-1-r+1)!} = n.\frac{(n-1)!}{(n-r)!}$

$= \frac{n.(n-1)!}{(n-r)!} = \frac{n!}{(n-r)!} = {}^{n}p_r = \text{L.H.S}$

ii. ${}^{n}P_r = {}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1}$

Sol: ${}^{n}P_r = {}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1}$

R.H.S $= {}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1}$

$= \frac{(n-1)!}{(n-1-r)!} + \frac{r.(n-1)!}{(n-1-r+1)!}$

$= \frac{(n-1)!}{(n-r-1)!} + \frac{r.(n-1)!}{(n-r)!}$

$= \frac{(n-1)!}{(n-r-1)!} + \frac{r.(n-1)!}{(n-r)(n-r-1)!}$

$= \frac{(n-1)!}{(n-r-1)!}\left[1 + \frac{r}{n-r}\right] = \frac{(n-1)!}{(n-r-1)!}\left[\frac{n-r+r}{n-r}\right]$

$= \frac{n(n-1)!}{(n-r)(n-r-1)!} = \frac{n!}{(n-r)!} = {}^{n}P_r = \text{L.H.S}$

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Q.4. **How many words can be formed from the letters of the following words using all letters when no letter is to be repeated?**

i. **PYTHON**

Sol: Total letters $= 6$, here n = 6, r = 6

Total words $^{6}P_{6} = \dfrac{6!}{(6-6)!} = 6!$

$= 6.5.4.3.2.1 = 720$

ii. **NETWORK**

Sol: Total letters $= 7$, here n = 7, r = 7

Total words $^{7}P_{7} = \dfrac{7!}{(7-7)!} = 7!$

$= 7.6.5.4.3.2.1 = 5040$

iii. **COMPUTER**

Sol: Total letters $= 8$, here n = 8, r = 8

Total words $^{8}P_{8} = \dfrac{8!}{(8-8)!} = 8!$

$= 8.7.6.5.4.3.2.1 = 40320$

Q.5. **How many signals can be given by 6 flags of different colors, using 2 flags at a time?**

Sol: Total flag $= 6 \Rightarrow n = 6$

Flag that can be used at a time $= 2 \Rightarrow r = 2$

Total signals $= {}^{n}P_{r} = {}^{6}P_{2} = \dfrac{6!}{(6-2)!}$

$= \dfrac{6.5.4.3.2.1}{4!} = \dfrac{6.5.\cancel{4!}}{\cancel{4!}} = 30$

Q.6. **How many signals can be given by 5 flags of different colors, when any number of flags are used at a time?**

Sol: Total flag $= 5$

When any number of flags is used at a time then

Total Signals $= {}^{5}P_{5} + {}^{5}P_{4} + {}^{5}P_{3} + {}^{5}P_{2} + {}^{5}P_{1}$ ____ (A)

$^{5}P_{5} = \dfrac{5!}{(5-5)!} = \dfrac{5.4.3.2.1}{0!} = 120$ ____ (1)

$^{5}P_{4} = \dfrac{5!}{(5-4)!} = \dfrac{5.4.3.2.1}{1!} = 120$ ____ (2)

$^{5}P_{3} = \dfrac{5!}{(5-3)!} = \dfrac{5.4.3.\cancel{2!}}{\cancel{2!}} = 60$ ____ (3)

$^{5}P_{2} = \dfrac{5!}{(5-2)!} = \dfrac{5.4.\cancel{3!}}{\cancel{3!}} = 20$ ____ (4)

$^{5}P_{1} = \dfrac{5!}{(5-1)!} = \dfrac{5.\cancel{4!}}{\cancel{4!}} = 5$ ____ (5)

Put (1), (2), (3), (4), (5) in (A)

$= 120 + 120 + 60 + 20 + 5 = 325$

Q.7. **How many 4 digit numbers can be formed, with distinct digits, with each digit odd?**

Sol: The odd digits are 1, 3, 5, 7, 9

Here n = 5, r = 4

So,

Total number of 4-digit distinct odd numbers

$^{5}P_{4} = \dfrac{5!}{(5-4)!} = 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

Q.8. **How many numbers between 100 and 1000 can be formed by using the digits 0,1,2,3,4,5 without repetition? How many of them are divisible by 5?**

Sol:

(a) Numbers between 100 and 1000 are 3 – digits numbers

Here n = 6, r = 3

Total 3-digit number

$^{6}P_{3} = \dfrac{6!}{(6-3)!} = \dfrac{6.5.4.\cancel{3!}}{\cancel{3!}} = 6 \times 5 \times 4 = 120$

When 0 at left $= {}^{5}P_{2} = \dfrac{5!}{(5-2)!} = \dfrac{5.4.\cancel{3!}}{\cancel{3!}} = 20$

Total 3 digit numbers between 100 and 1000

$= 120 - 20 = 100$

(b) A numbers Divisible by '5' if its units digit is either 0 or 5

When 0 at unit place

$= {}^{5}P_{2} = \dfrac{5!}{(5-2)!} = \dfrac{5.4.\cancel{3!}}{\cancel{3!}} = 20$

When 0 at left $= {}^{4}P_{1} = \dfrac{4!}{(4-1)!} = \dfrac{4.\cancel{3!}}{\cancel{3!}} = 4$

Total 3 digit number when 0 at unit place

$= 20 - 4 = 16$

When 5 at unit place

$= {}^{5}P_{2} = \dfrac{5!}{(5-2)!} = \dfrac{5.4.\cancel{3!}}{\cancel{3!}} = 20$

Total numbers when 0 and 5 at unit place

$= 20 + 16 = 36$

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Q.9. **Find the numbers greater than 35000 that can be formed from the digits 1,2,3,4,5,6, without repeating any digit.**

Sol: $Digits = 1,2,3,4,5,6$

$Total\ digits = 6$

There are two options for number greater than 35000

i) If first digit has 4,5,6 all combinations with be greater than 35,000.

$Total\ digit = 4 \times 5 \times 6 = 120$

There are 3 options for first digit options so

$Total\ numbers = 120 \times 3 = 360$

ii) If first digit is 3

the second must be 5 or 6

$Total\ combinations = 4! + 4! = 24 + 24 = 48$

Total numbers greater than

$35000 = 360 + 48 = 408$

Q.10. **Find the number of 5-digit numbers that can be formed from the digits 1,2,4,6,8 (when no digit is repeated), but**

i. **The digit 2 and 8 are next to each other's**

ii. **The digits 2 and 8 are not next to each other.**

Sol: $Digits = 1,2,4,6,8$

$Total\ Digits = 5$

$Total\ arrangements = 5! = 5.4.3.2.1 = 120$

i. The digits 2 and 8 are when 2 and 8 are next to each other we consider it as a single unit '28' or '82'

Then, $Total\ Digits = 4$

$Total\ numbers = 2 \times 4! = 2 \times 24 = 48$

ii. The digits 2 and 8 are not next to each other $= 120 - 48 = 72$

Q.11. **How many 6-digit numbers can be formed without repeating any digit from the digits 0,1,2,3,4,5? In how many of them will 0 be at the tens place?**

Sol: $Digits = 0,1,2,3,4,5$

$Total\ 6\ Digits = 6!$

$= 6.5.4.3.2.1 = 720$

When '0' fixed at tens place then total digits

Becomes $= 5! = 5.4.3.2.1 = 120$

Total 6 digit numbers

$= 720 - 120 = 600$

Q.12. **How many 5-digit multiples of 5 can be formed from the digits 2,3,5,7,9, when no digit is repeated.**

Sol: $Digits = 2,3,5,7,9$

Total five digits

For multiple of '5'

5 is fixed at units place

$\Rightarrow Total\ digit = 4$

Then, $Total\ multiples\ of\ 5 = 4! = 4.3.2.1 = 24$

Q.13. **In how many ways can 8 different books including 2 on English be arranged on a shelf in such a way that the English books are never together?**

Sol: Suppose,

E_1, E_2 are two English books and

$B_1, B_2, B_3, B_4, B_5, B_6$ remaining books.

$$\text{Total} = {}^8P_8 = \frac{8!}{(8-8)!} = \frac{8.7.6.5.4.3.2.1}{0!} = 40320$$

English books are together

Case I $B_1, B_2, B_3, B_4, B_5, B_6, \boxed{E_1E_2}$

$$= {}^7P_7 = \frac{7!}{(7-7)!} = \frac{7.6.5.4.3.2.1}{0!} = 5040$$

Case II $B_1, B_2, B_3, B_4, B_5, B_6, \boxed{E_2E_1}$

$$= {}^7P_7 = 5040$$

English books together $= 5040 + 5040 = 1080$

English books are not together = Total − Together

$= 40320 - 10080 = 30240$

نوٹ:

E_1E_2 کو ملا کر ایک کتاب بنالیں اس میں دو آپشن بنیں گے۔ E_1 کو پہلے اور E_2 بعد اور E_2 کو پہلے اور E_1 کو بعد میں رکھیں۔

Q.14 **Find the number of arrangements of 3 different books on English and 5 different books on Urdu for placing them on a shelf such that the books on the same subject are together.**

Sol: E_1, E_2, E_3 are English and U_1, U_2, U_3, U_4, U_5 Urdu books.

Case I $E_1, E_2, E_3 \times U_1, U_2, U_3, U_4, U_5$

$= {}^3P_3 \times {}^5P_5$

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$$= \frac{3!}{(3-3)!} \times \frac{5!}{(5-5)!} = 3.2.1 \times 5.4.3.2.1$$

$$= 6 \times 120 = 720$$

Case II $U_1, U_2, U_3, U_4, U_5 \times E_1, E_2, E_3$

$$= {}^5P_5 \times {}^3P_3 = \frac{5!}{(5-5)!} \times \frac{3!}{(3-3)!}$$

$$= \frac{5.4.3.2.1}{0!} \times \frac{3.2.1}{0!}$$

$$= 120 \times 6 = 720$$

Total $= 720 + 720 = 1440$

Q.15 **In how many ways can 5 boys and 4 girls be seated on a bench so that the girls and the boys occupy alternate seats?**

Sol: B represents boys and G girls.

Total number of ways:

$$= {}^5P_5 \times {}^4P_4$$

$$= \frac{5!}{(5-5)!} \times \frac{4!}{(4-4)!} = \frac{5.4.3.2.1}{0!} \times \frac{4.3.2.1}{0!}$$

$$= 120 \times 24 = 2880$$

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